

Ethanol Blending in Gasoline – France

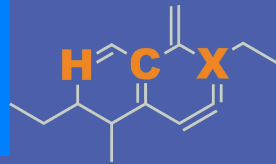
July 2026



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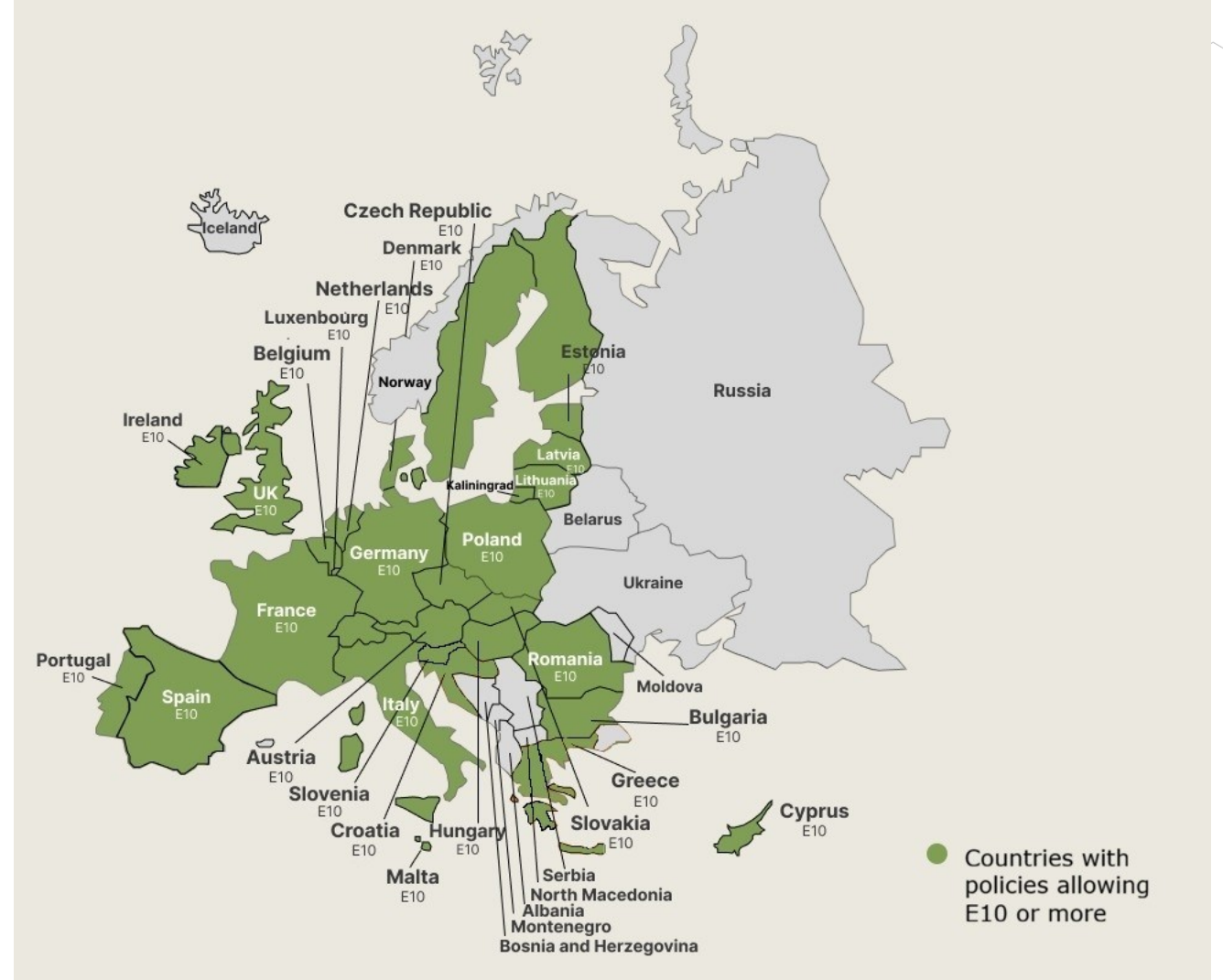


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Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

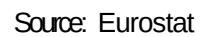
Ethanol Gasoline Blending in France



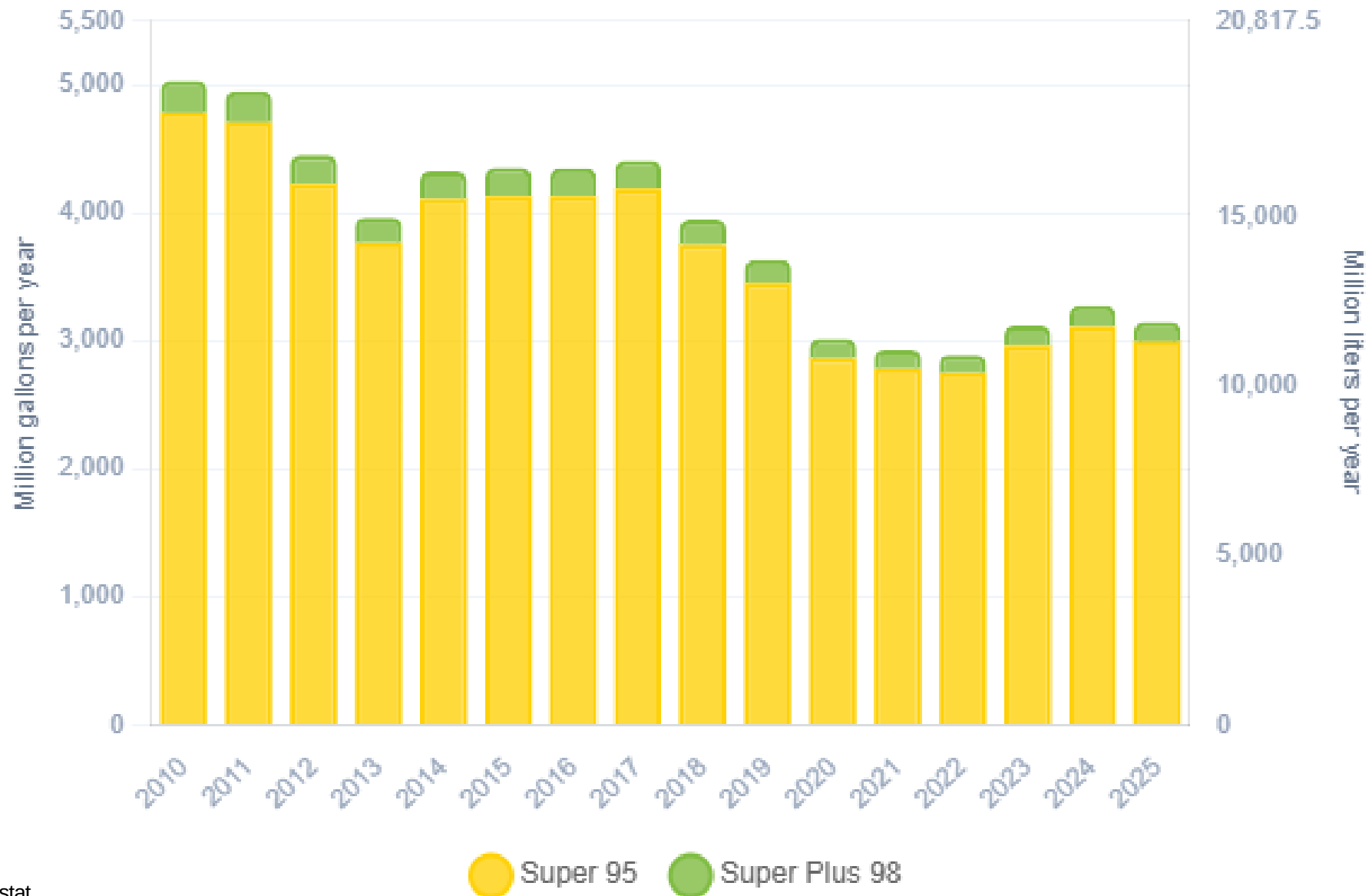
In 2025, gasoline consumption in France was 14,702 million liters (3,884 million gallons) and growth rate was 1.5%. Gasoline production and net imports were 11,926 and 1,072 million liters (3,150 and 283 million gallons) correspondingly. France complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.7 RON.

In 2025, ethanol consumption in France was 1,518 million liters (401 million gallons) representing 10% of gasoline demand. Ethanol production and Net Imports were 1,192 and 325 million liters (315 and 86 million gallons) correspondingly. Ethanol sources are Corn 32%, Cereals 21% and Sugar Beet 47%.

The FARO90 logo is shown in a blue box, with the text "FARO90" in white. To the right of the logo is a chemical structure diagram of a polyene chain, featuring a series of conjugated double bonds and a terminal group labeled "X".

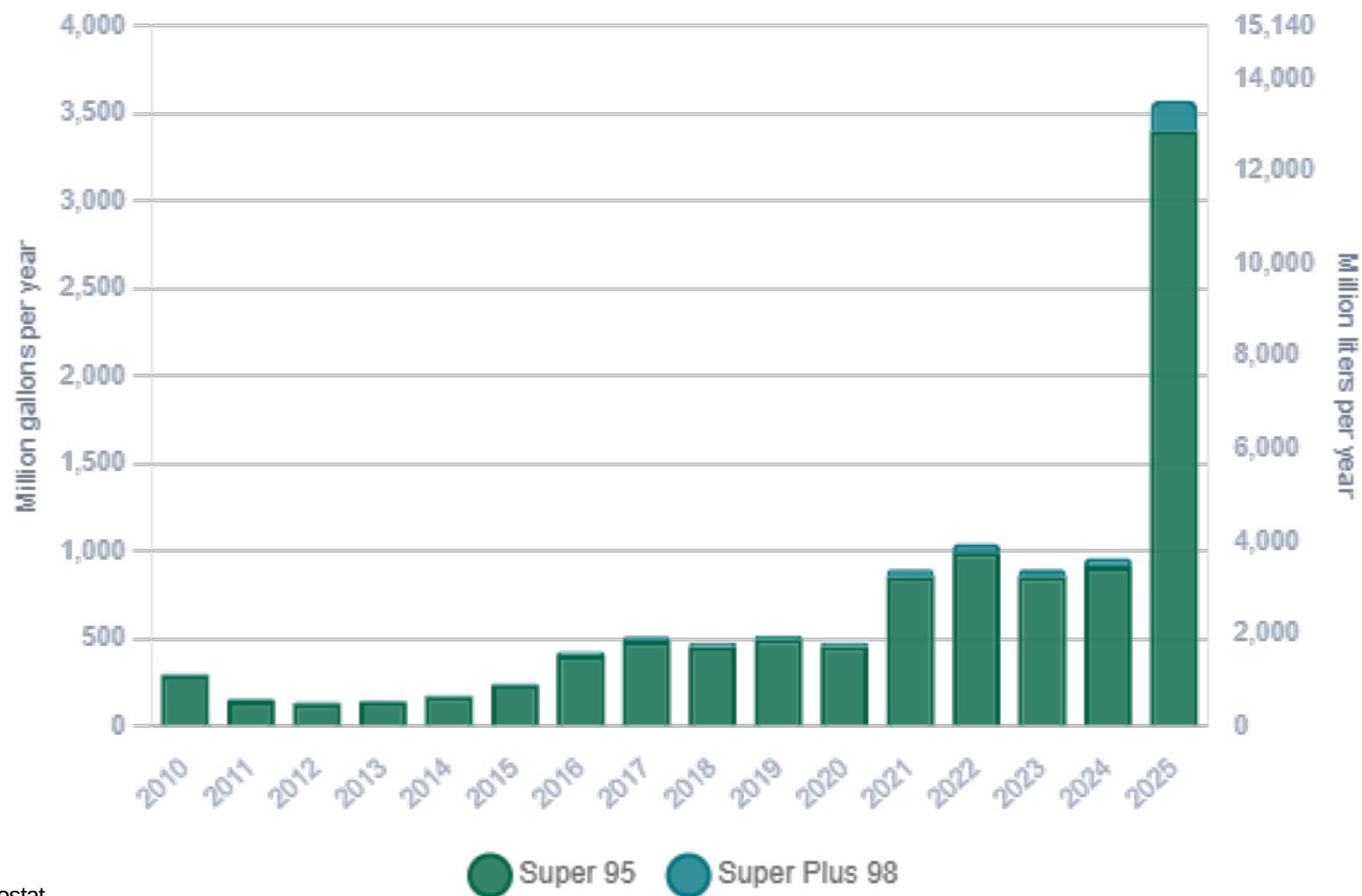


Gasoline Production in France



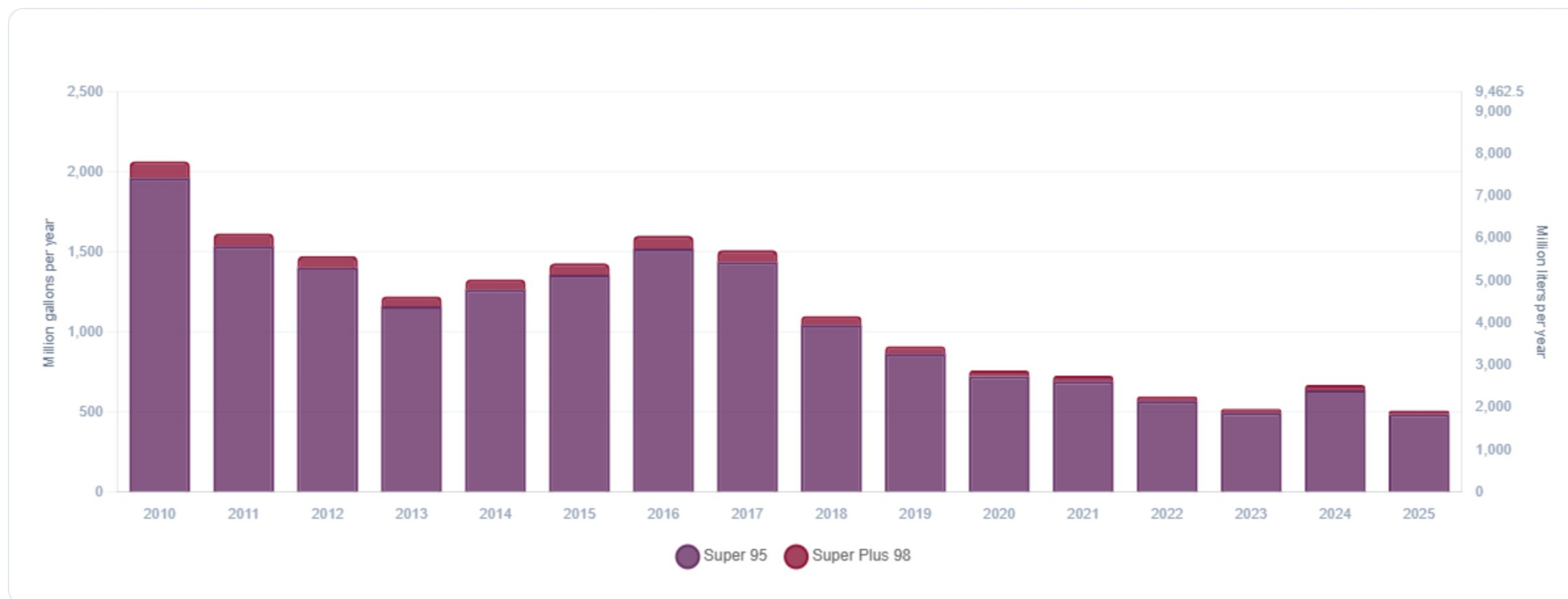
Source: Eurostat

Gasoline Imports in France



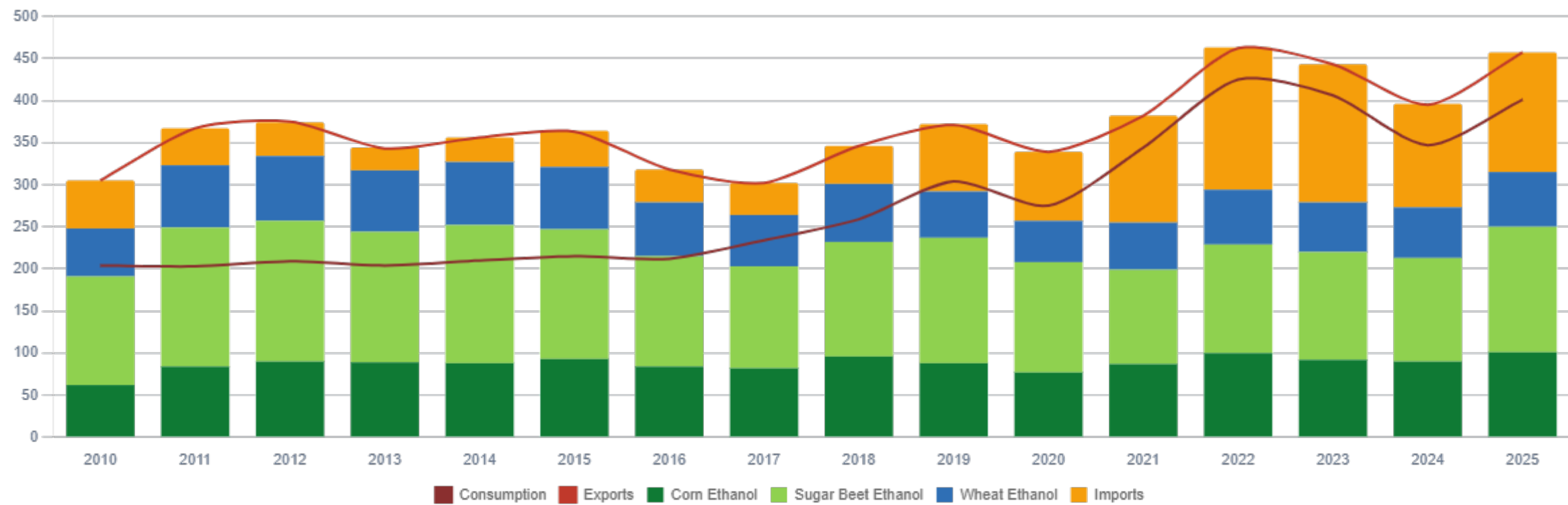
Source: Eurostat

Gasoline Exports in France



Source: Eurostat

Ethanol Balance in France



Source: Eurostat

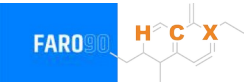
Gasoline Quality in France

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.2							
Bioethanol in Gasoline Sales (%cal)	8.6							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

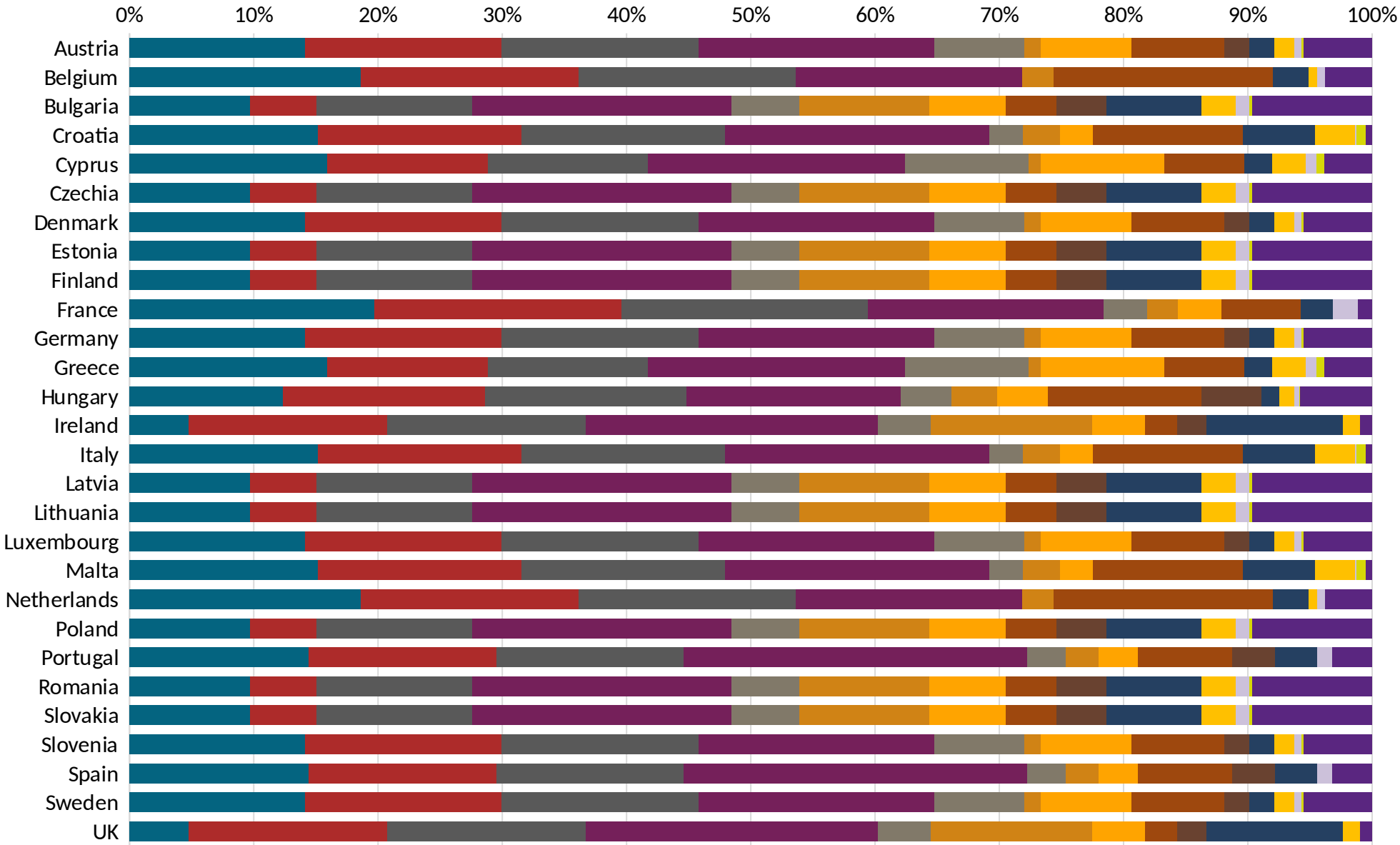
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



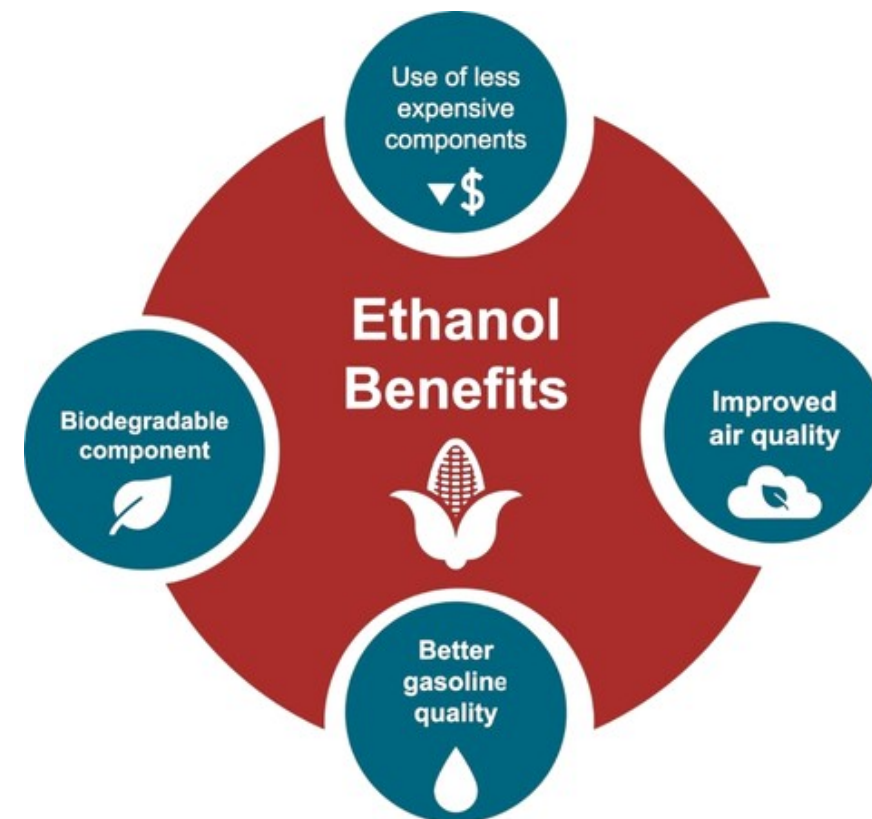
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

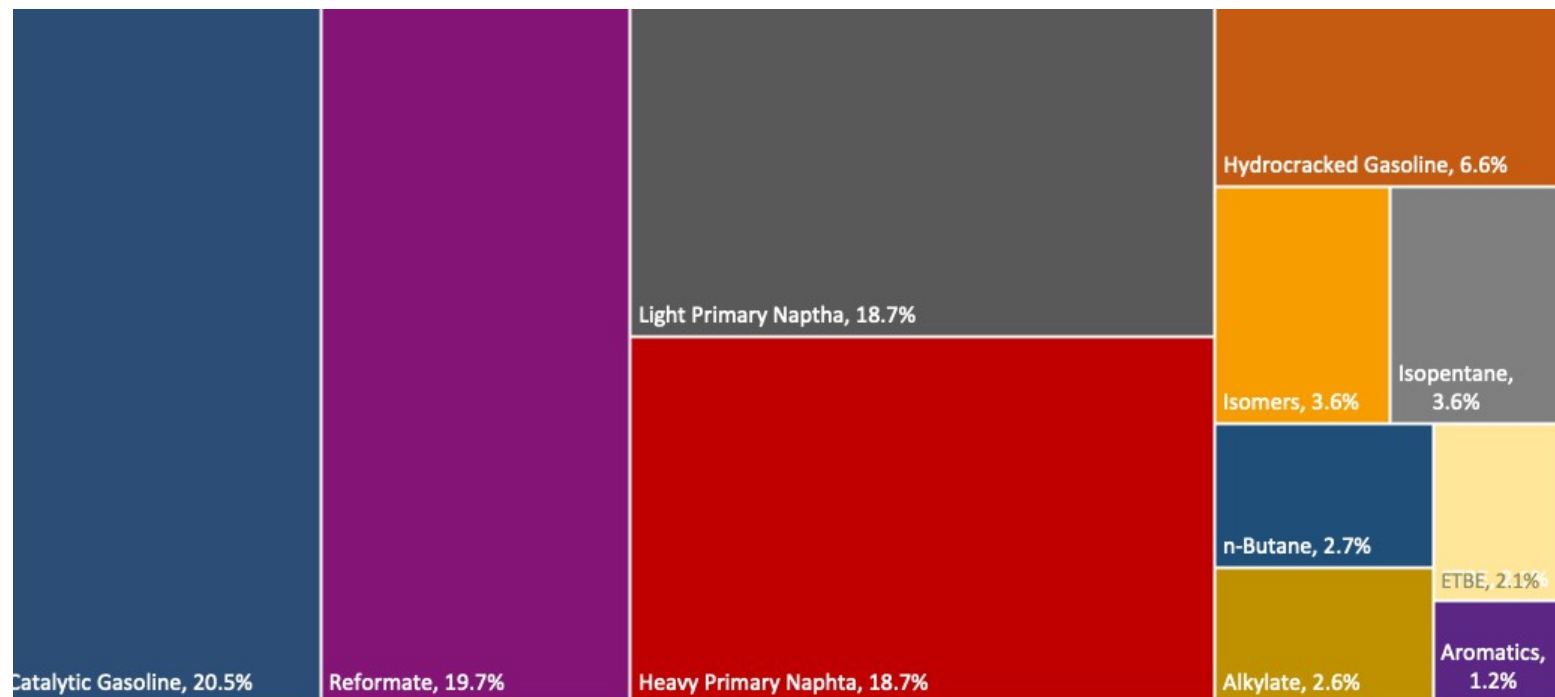
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

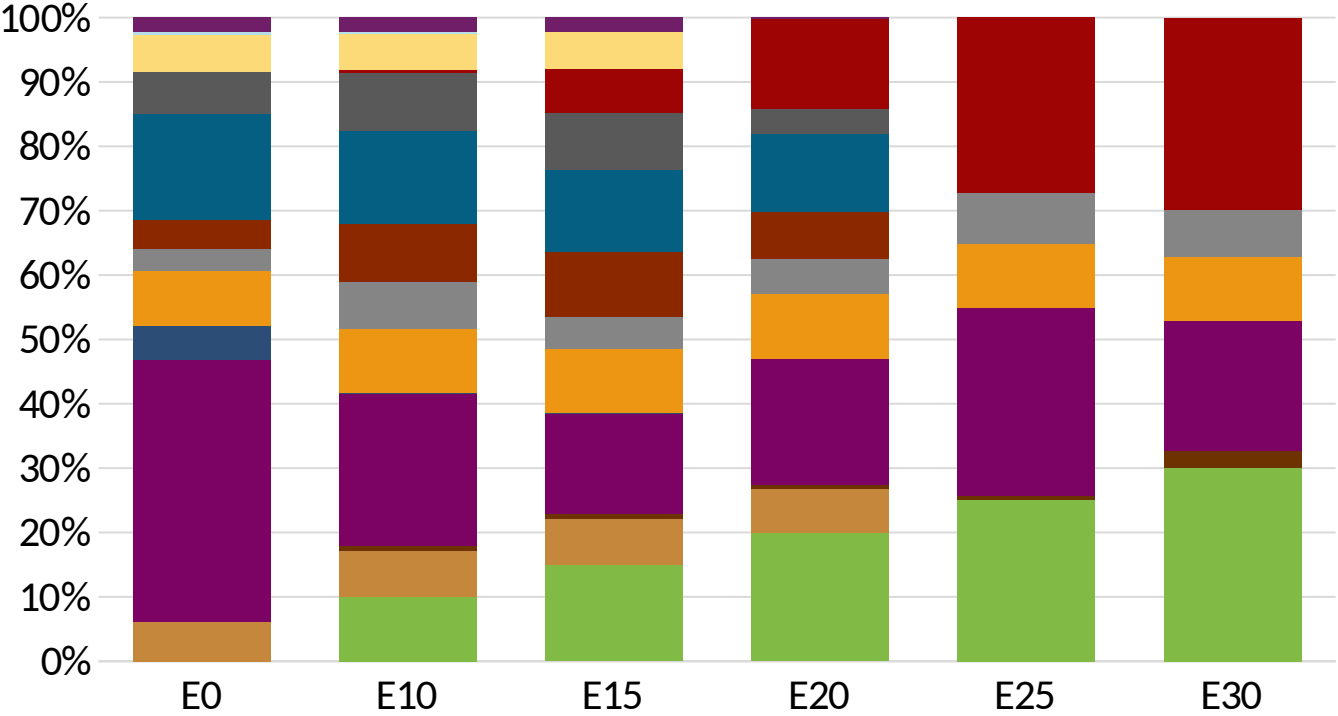
The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



France Available Blending Components

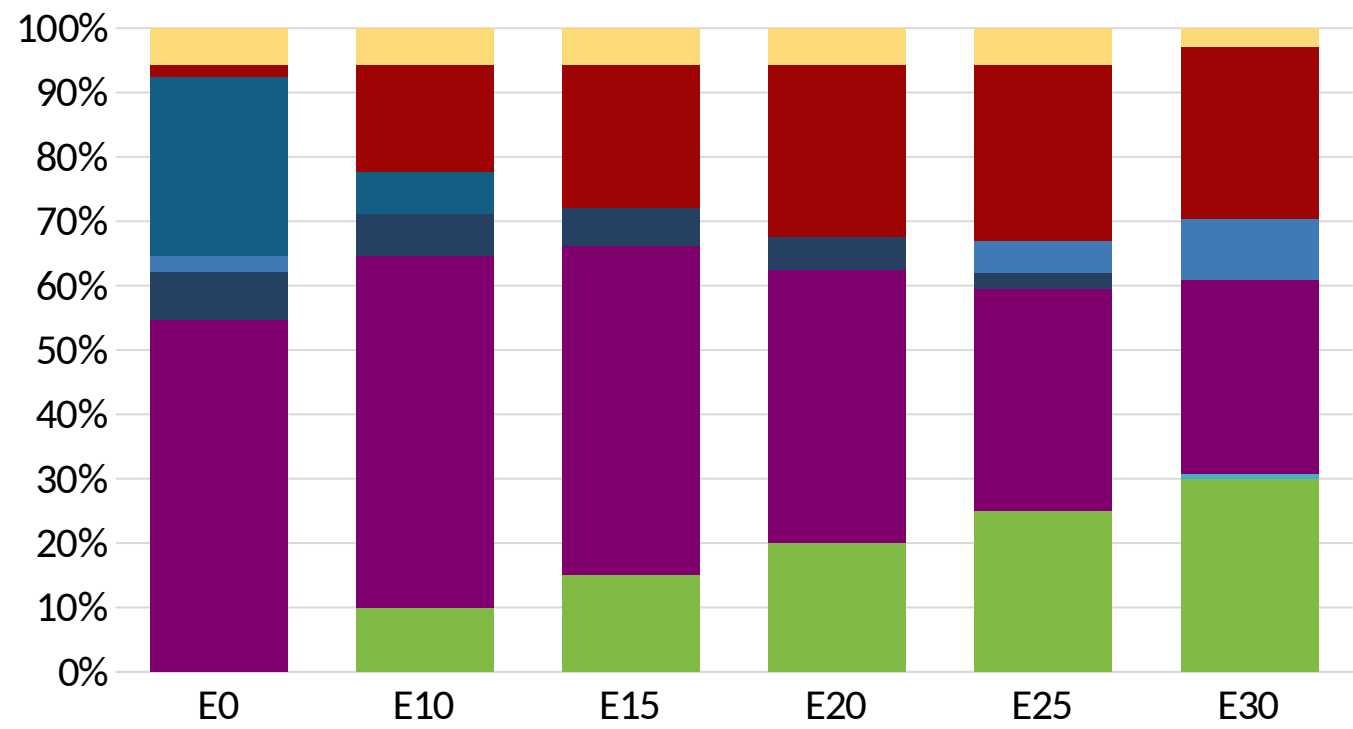


France – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

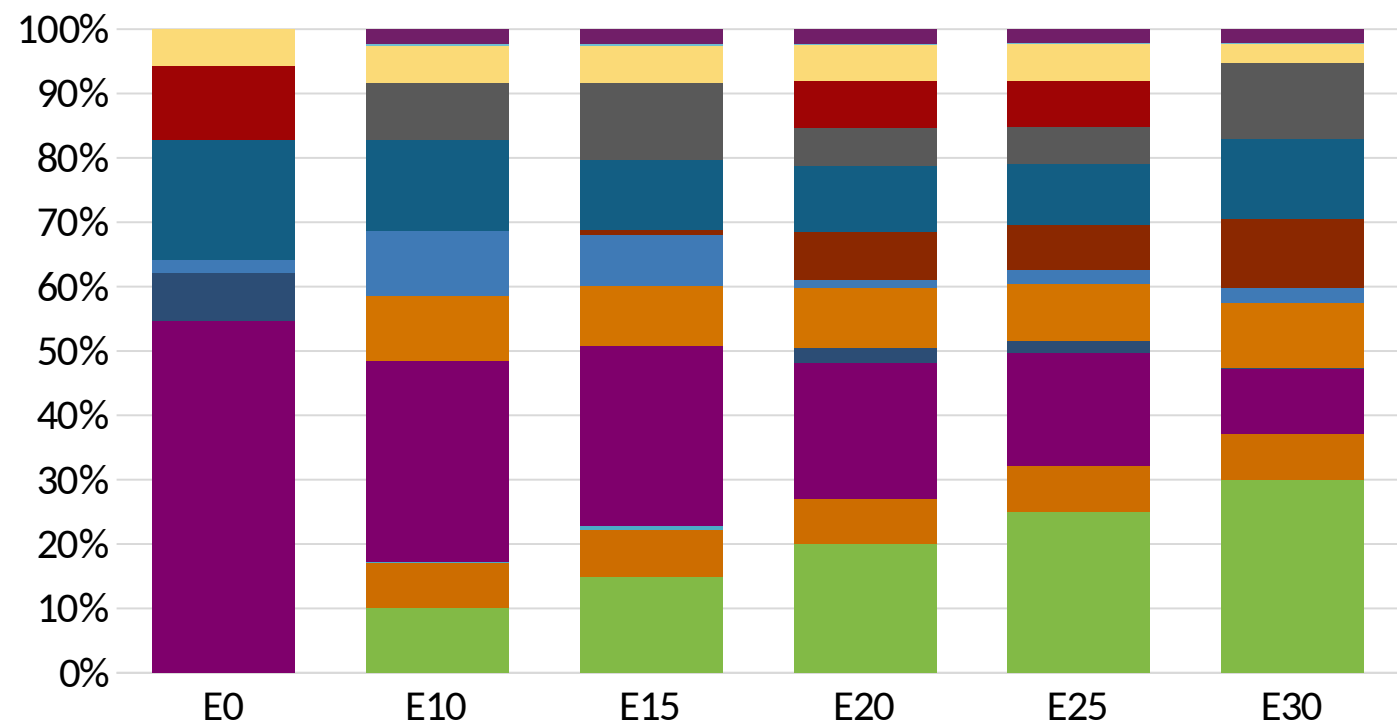
France - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.143	\$ 2.019	\$ 1.948	\$ 1.873	\$ 1.795	\$ 1.721

Average CIF 2025 Prices

France – Gasoline 95 – Increased Octane

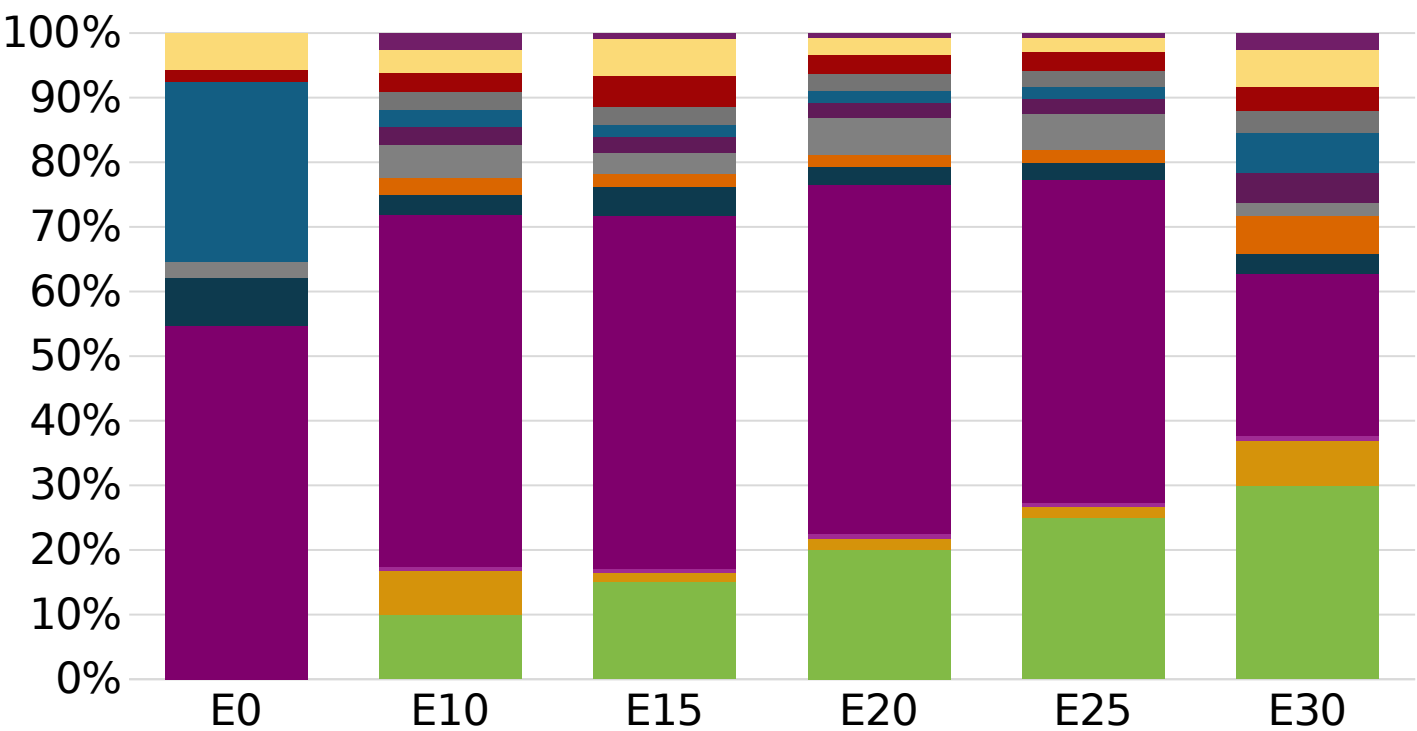


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	101.7
Price (US\$/gal)	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009

Source:
Faro90

France – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.3	101.4	102.4	103.7	104.6
Price (US\$/gal)	\$ 2.143	\$ 2.143	\$ 2.143	\$ 2.143	\$ 2.143	\$ 2.143

Source:
Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

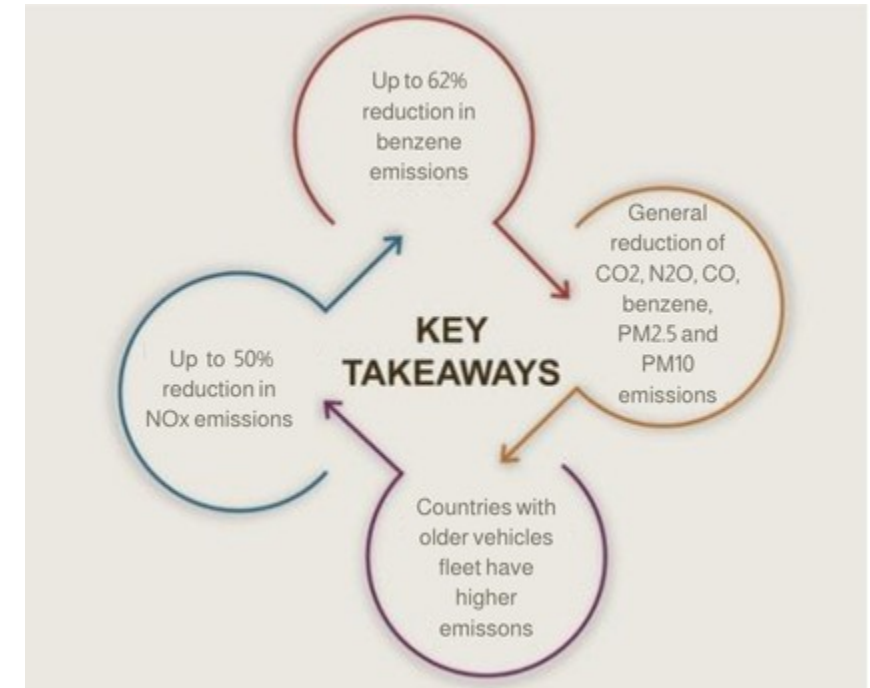
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

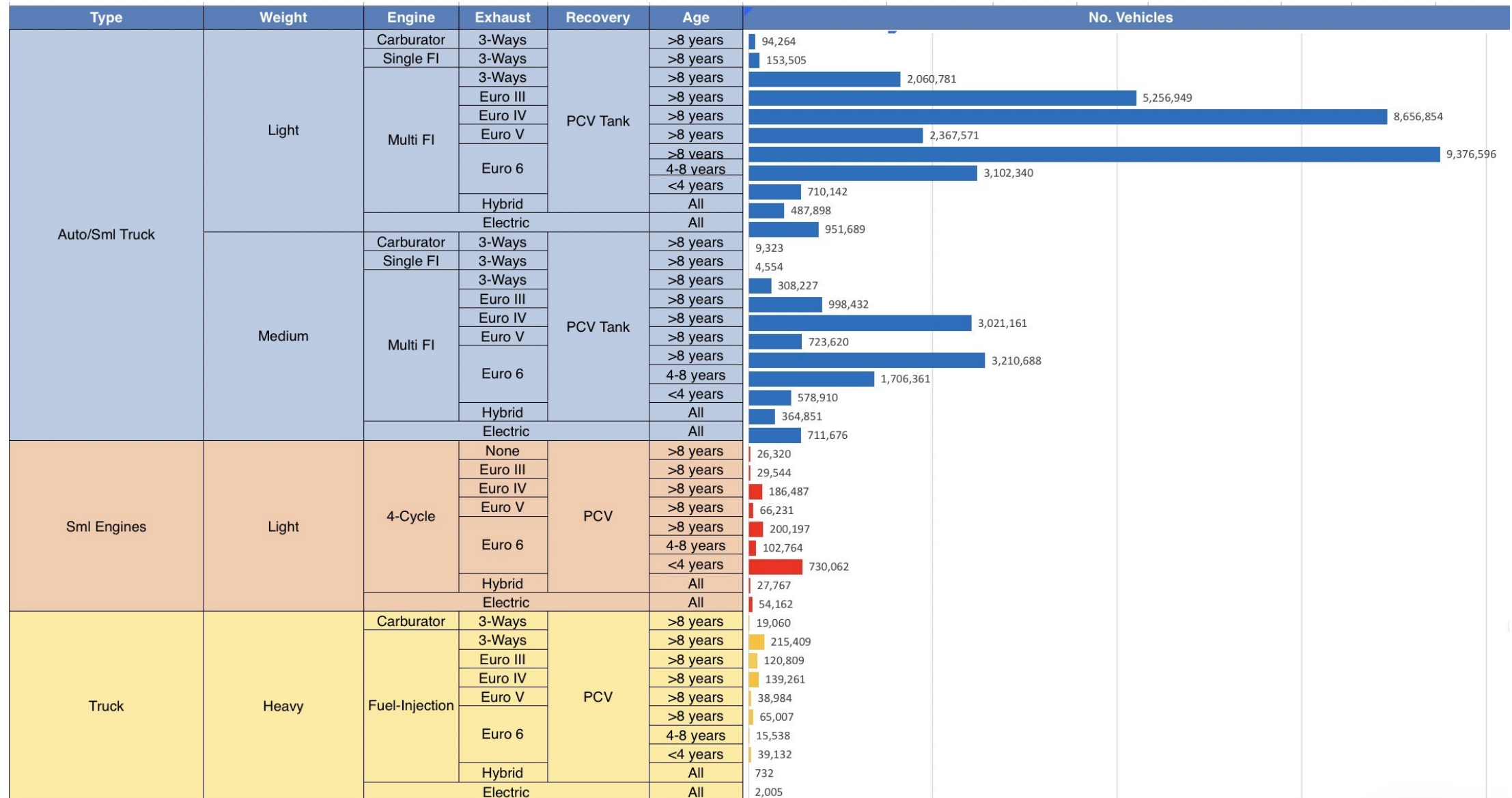
Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - France

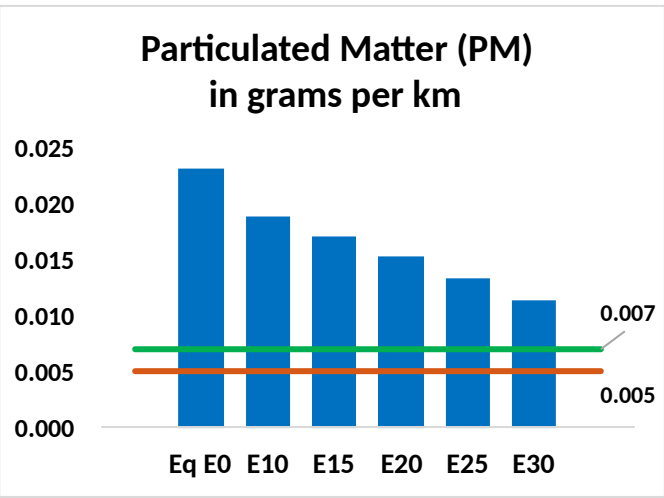
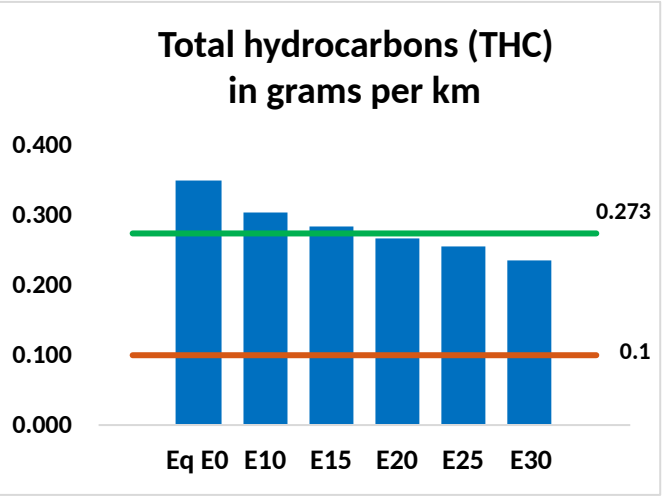
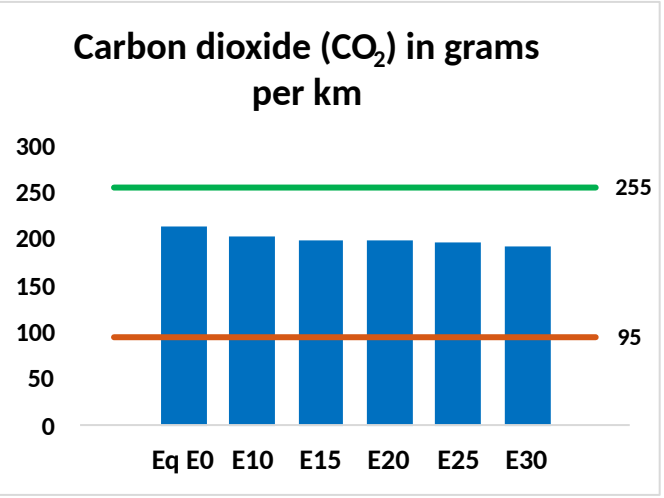
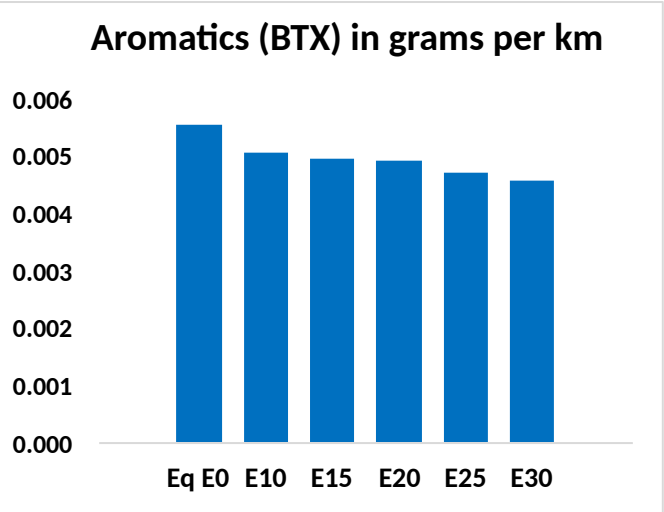
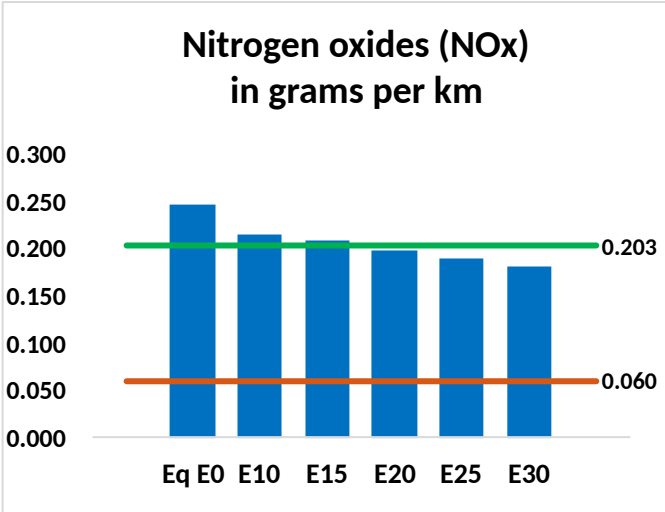
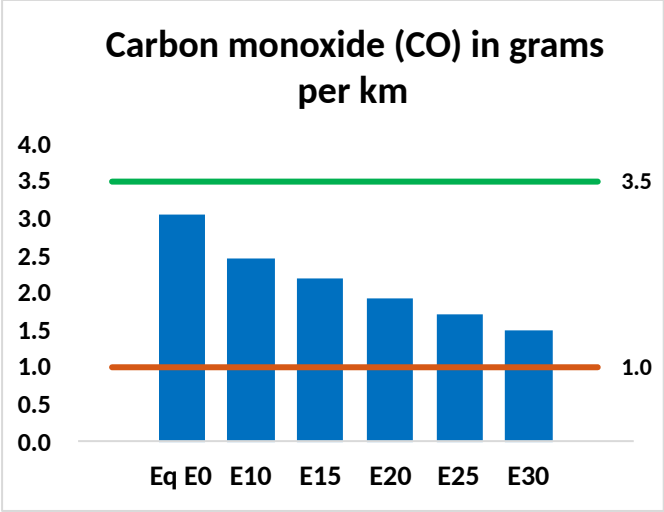
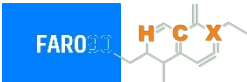


Vehicle Fleet: 46,878,458
 Gasoline Vehicle Fleet:
 22,915,839

Source: Eurostat, ACEA

France – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard



France – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,829,288	1,646,298	1,463,308	1,305,089	1,142,065	1,015,561	884,729	-20%	-38%
CO2	128,233,648	124,936,211	121,821,966	119,372,703	118,160,712	117,444,650	115,279,808	-5%	-8%
VOC	208,099	194,530	180,961	170,315	159,820	151,816	140,903	-13%	-23%
NOx	147,567	136,254	128,697	124,955	118,807	114,044	108,400	-13%	-19%
SOx	1,484	1,372	1,260	1,165	1,074	975	871	-15%	-28%
NH3	44,156	43,719	43,282	43,487	43,976	44,320	44,606	-2%	0%
N2O	1,711	1,702	1,695	1,703	1,738	1,758	1,778	-1%	2%
CH4	44,539	44,539	44,539	45,207	46,187	47,033	47,835	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,783	1,709	1,634	1,586	1,542	1,511	1,463	-8%	-13%
Acetaldehyde	4,009	4,602	6,752	10,282	14,009	16,007	18,915	68%	249%
Formaldehyde	13,095	12,732	14,841	17,427	18,258	20,198	21,987	13%	39%
Benzene	3,327	3,197	3,029	2,981	2,957	2,828	2,736	-9%	-11%
PM2.5	7,507	6,825	6,142	5,542	4,954	4,324	3,665	-18%	-34%
PM10	6,379	5,799	5,219	4,709	4,209	3,674	3,114	-18%	-34%

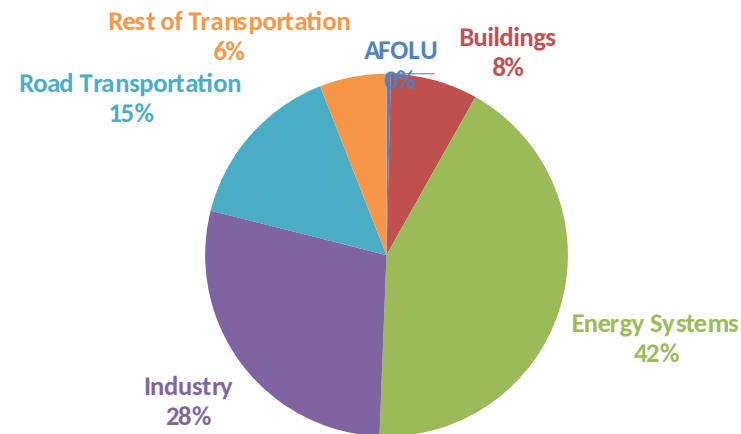
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

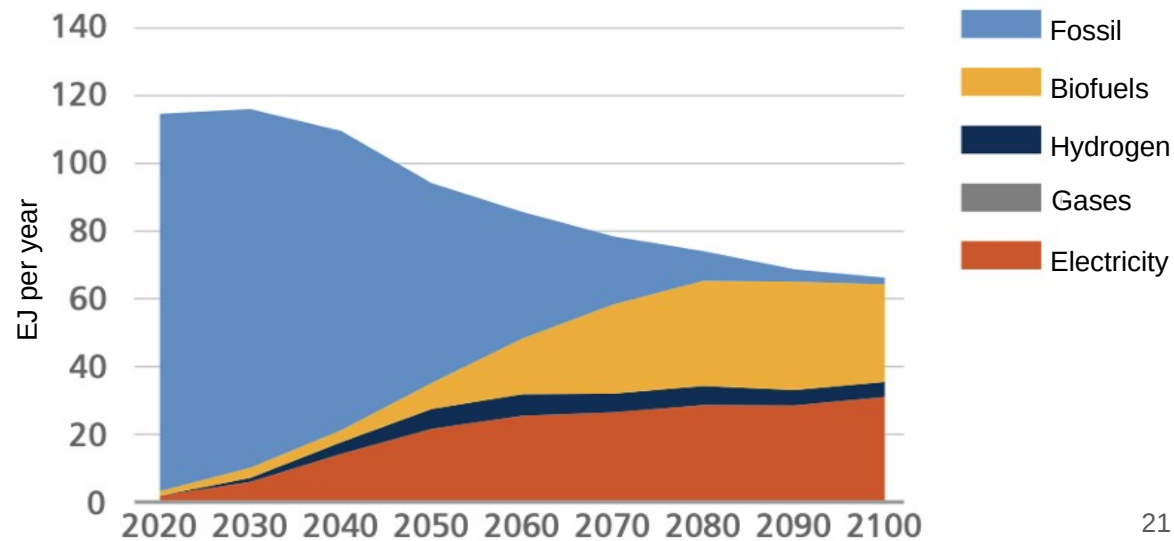
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

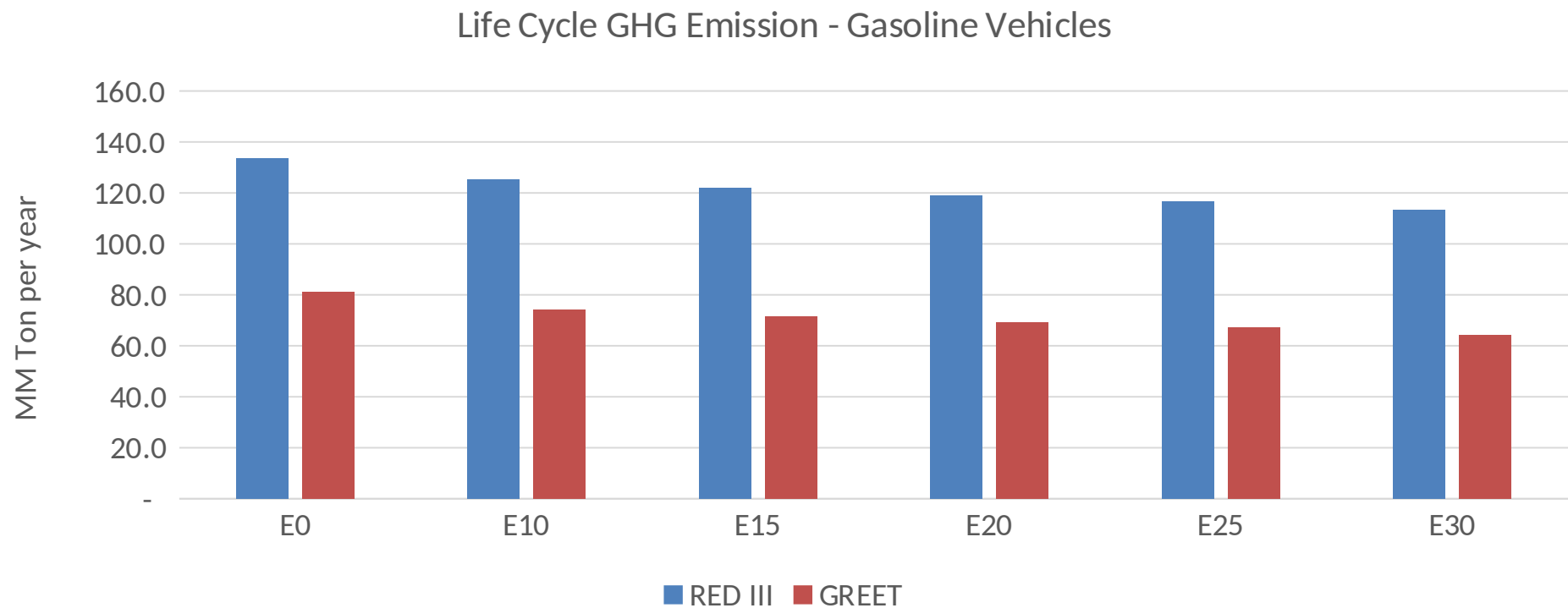
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



France – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	331.1
Target @ 2035 (MMT/year)	186.4
Delta	144.7